

MIMIC-III NLP Assignment

Comparing spaCy, scispaCy, and medspaCy on Real ICU Discharge Notes

Why Compare Three NLP Tools on Clinical Text?

The Problem It Solves

General NLP tools miss a lot of clinical meaning - they don't know 'MI' means heart attack, or that 'denies pain' is a negative finding.

What Makes It Different

spaCy is general-purpose. scispaCy adds biomedical training. medspaCy adds clinical context - negation, family history, hypothetical statements.

Why This Matters

Choosing the right NLP tool changes what information you can reliably pull out of real patient notes.

How the Pipeline Is Structured

Cohort & Data Source

- Data pulled from Google BigQuery (MIMIC-III public dataset)
- Cohort: patients with ICD-9 code 430 (subarachnoid hemorrhage)
- Note type: Discharge summaries only
- 888 matching notes found; a smaller sample used for processing speed

What Each Tool Does

- Entity extraction: find clinical terms in the text
- word2vec: turn words into number vectors based on context
- t-SNE: compress those vectors down to 2D so we can plot and compare them
- Same steps run for spaCy, scispaCy, and medspaCy on the same notes

spaCy — General-Purpose NLP

What it does:

Runs standard named entity recognition (people, places, dates, numbers) plus word2vec and t-SNE on the same notes.

Why it matters:

It's a fair baseline to measure the other two tools against - free of any medical training.

How we ran it:

Load en_core_web_sm, extract entities, tokenize, train word2vec, plot tuned t-SNE.

Finding

Top entity types were CARDINAL, ORG, PERSON, and DATE - generic categories. It missed almost all clinical concepts (diagnoses, symptoms, medications).

Word2vec vocabulary size: 2,166 words

scispaCy — Biomedical NLP

What it does:

Uses models trained on biomedical literature to recognize diseases, chemicals, and other domain-specific terms.

Why it matters:

It understands clinical vocabulary that general spaCy simply doesn't recognize.

How we ran it:

Load `en_core_sci_sm` and `en_ner_bc5cdr_md`, extract entities, tokenize, train word2vec, plot tuned t-SNE.

Finding

Found 1,178 DISEASE entities and 678 CHEMICAL entities - a much more clinically useful breakdown than spaCy's generic categories.

Word2vec vocabulary size: 2,321 words

medspaCy — Clinical Context Detection

Adds negation, family history, and hypothetical detection on top of entities

What it does:

Flags whether a clinical concept is negated, historical, hypothetical, or about a family member - not just whether it's mentioned.

Why it matters:

'Denies chest pain' and 'has chest pain' are opposite findings. Only medspaCy tells them apart.

How we ran it:

Load medspaCy, add target rules for SAH-related concepts, run context detection over the sample.

Important finding

Unlike spaCy and scispaCy, medspaCy's entity finder is rule-based and starts empty - it only finds concepts we explicitly define with target rules.

Word2vec vocabulary size: 2,223 words

RESULTS

Side-by-Side Comparison

	spaCy	scispaCy	medspaCy
Entity focus	General (people, orgs, dates)	Biomedical (disease, chemical)	Clinical context (negation, etc.)
Top categories found	CARDINAL, ORG, PERSON	DISEASE, CHEMICAL	Rule-defined only
Needs custom setup?	No	No	Yes - target rules required
word2vec vocab size	2,166	2,321	2,223
Best for	Quick, general text	Clinical concept extraction	Negation & context accuracy

From Entities to Clinical Insight

Key Takeaways

- General NLP misses most clinical meaning in notes
- Domain training (scispaCy) surfaces real clinical concepts
- Context detection (medspaCy) tells a positive finding from a negative one
- Picking the right tool changes what you can reliably extract from notes

How to Reproduce This Analysis

- Get PhysioNet access to MIMIC-III
- Load MIMIC-III into Google BigQuery
- Run the cohort query in the notebook
- Install spaCy, scispaCy, and medspaCy
- Run entity extraction, word2vec, and t-SNE for each tool